KAIZEN MATHS

Lesson starter

Continuous Random Variables and Density Functions

Expectation & Variance

Name

S1

The density of X is $f(x) = \frac{1}{18}x$ for $0 \le x \le 6$, and 0 otherwise.Find E(X).

S2

The density of X is $f(x) = \frac{1}{18}x$ for $0 \le x \le 6$, and 0 otherwise.Find Var(X) and the standard deviation.

S3

The density of X is $f(x) = \frac{1}{18}x$ for $0 \le x \le 6$, and 0 otherwise.Find the lower quartile of X.

S4

The density of X is $f(x) = \frac{3}{125}x^2$ for $0 \le x \le 5$, and 0 otherwise.Find E(X) and Var(X).

Show your working

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Exit ticket

Continuous Random Variables and Density Functions

Expectation & Variance

Name

E1

The density of X is f(x) = $\frac{2}{81}$x for 0 ≤ x ≤ 9, and 0 otherwise.Find E(X).

E2

The density of X is f(x) = $\frac{1}{72}$x for 0 ≤ x ≤ 12, and 0 otherwise.Find Var(X) and the standard deviation.

E3

The density of X is f(x) = $\frac{1}{50}$x for 0 ≤ x ≤ 10, and 0 otherwise.Find the upper quartile of X.

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Teacher answers & working

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S1

The density of X is f(x) = $\frac{1}{18}$x for 0 ≤ x ≤ 6, and 0 otherwise.Find E(X).

Answer:

$E(X) = 4$

Working:

Expected value is the weighted average, so multiply x by the density before integrating.

Use E(X) = ∫ from 0 to 6 of x f(x) dx.

Substitute f(x): E(X) = ∫ from 0 to 6 of x($\frac{1}{18}$x) dx.

Multiply inside the integral: x($\frac{1}{18}$x) = $\frac{1}{18}$x².

Integrate x². E(X) = $\frac{1}{18}$[$\frac{x^3}{3}$] from 0 to 6.

Substitute the limits: E(X) = $\frac{1}{18}$($\frac{216}{3}$ - $\frac{0^3}{3}$).

Simplify: E(X) = $\frac{2(216)}{3\times36}$ = 4.

$E(X) = 4 \approx 4.000$

S2

The density of X is f(x) = $\frac{1}{18}$x for 0 ≤ x ≤ 6, and 0 otherwise.Find Var(X) and the standard deviation.

Answer:

$\mathrm{Var}(X) = 2,\ \mathrm{SD} = 1.414$

Working:

Variance needs both E(X) and E(X²).

From E(X) = ∫ x f(x) dx, E(X) = 4.

Now use E(X²) = ∫ from 0 to 6 of x² f(x) dx.

Substitute f(x): E(X²) = ∫ from 0 to 6 of x²($\frac{1}{18}$x) dx.

Multiply inside the integral: x²($\frac{1}{18}$x) = $\frac{1}{18}$x³.

Integrate x³. E(X²) = $\frac{1}{18}$[$\frac{x^4}{4}$] from 0 to 6.

Substitute the limits: E(X²) = $\frac{1}{18}$($\frac{1296}{4}$ - $\frac{0^4}{4}$) = 18.

Use $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

$\mathrm{Var}(X) = 18 - (4)^2 = 2$.

SD(X) = sqrt(2) ≈ 1.414.

$\mathrm{Var}(X) = 2$

Teacher answers & working

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S3

The density of X is f(x) = $\frac{1}{18}$x for 0 ≤ x ≤ 6, and 0 otherwise.Find the lower quartile of X.

Answer:

3.000

Working:

Use the CDF, because quartiles are found from cumulative probability.

The density integrates to F(x) = ∫ from 0 to x of ($\frac{1}{18}$)t dt = $\frac{x^2}{36}$ for 0 ≤ x ≤ 6.

The lower quartile is the value q where F(q) = 0.25.

$\frac{q^2}{36} = 0.25$.

$q^2 = 9$.

q = sqrt(9) ≈ 3.000.

lower quartile ≈ 3.000

S4

The density of X is f(x) = $\frac{3}{125}$x² for 0 ≤ x ≤ 5, and 0 otherwise.Find E(X) and Var(X).

Answer:

$E(X) = \frac{15}{4}$, $Var(X) = \frac{15}{16}$

Working:

Use E(X) = ∫ x f(x) dx over the support.

Substitute f(x): E(X) = ∫ from 0 to 5 of x($\frac{3}{125}$x²) dx.

Multiply inside the integral: x($\frac{3}{125}$x²) = $\frac{3}{125}$x³.

Integrate x³: E(X) = $\frac{3}{125}$[$\frac{x^4}{4}$] from 0 to 5.

Substitute the limits: E(X) = $\frac{3}{125}$($\frac{625}{4}$ - $\frac{0^4}{4}$) = $\frac{15}{4}$.

Find E(X²) before variance.

Use E(X²) = ∫ from 0 to 5 of x² f(x) dx.

Substitute and multiply: E(X²) = ∫ from 0 to 5 of x²($\frac{3}{125}$x²) dx = $\frac{3}{125}$∫ from 0 to 5 of x⁴ dx.

Integrate x⁴: E(X²) = $\frac{3}{125}$[$\frac{x\boxtimes}{5}$] from 0 to 5 = 15.

$Var(X) = E(X^2) - [E(X)]^2 = 15 - (\frac{15}{4})^2$.

$Var(X) = \frac{15}{16} \approx 0.938$

Teacher answers & working

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E1

The density of X is f(x) = $\frac{2}{81}$x for 0 ≤ x ≤ 9, and 0 otherwise.Find E(X).

Answer:

$E(X) = 6$

Working:

Expected value is the weighted average, so multiply x by the density before integrating.

Use E(X) = ∫ from 0 to 9 of x f(x) dx.

Substitute f(x): E(X) = ∫ from 0 to 9 of x($\frac{2}{81}$x) dx.

Multiply inside the integral: x($\frac{2}{81}$x) = $\frac{2}{81}$x².

Integrate x². E(X) = $\frac{2}{81}$[$\frac{x^3}{3}$] from 0 to 9.

Substitute the limits: E(X) = $\frac{2}{81}$($\frac{729}{3}$ - $\frac{0^3}{3}$).

Simplify: E(X) = $\frac{2(729)}{3\times81}$ = 6.

$E(X) = 6 \approx 6.000$

E2

The density of X is f(x) = $\frac{1}{72}$x for 0 ≤ x ≤ 12, and 0 otherwise.Find Var(X) and the standard deviation.

Answer:

$\mathrm{Var}(X) = 8,\ \mathrm{SD} = 2.828$

Working:

Variance needs both E(X) and E(X²).

From E(X) = ∫ x f(x) dx, E(X) = 8.

Now use E(X²) = ∫ from 0 to 12 of x² f(x) dx.

Substitute f(x): E(X²) = ∫ from 0 to 12 of x²($\frac{1}{72}$x) dx.

Multiply inside the integral: x²($\frac{1}{72}$x) = $\frac{1}{72}$x³.

Integrate x³. E(X²) = $\frac{1}{72}$[$\frac{x^4}{4}$] from 0 to 12.

Substitute the limits: E(X²) = $\frac{1}{72}$($\frac{20736}{4}$ - $\frac{0^4}{4}$) = 72.

Use $\mathrm{Var}(X) = E(X^2) - [E(X)]^2$.

$\mathrm{Var}(X) = 72 - (8)^2 = 8$.

SD(X) = sqrt(8) ≈ 2.828.

$\mathrm{Var}(X) = 8$

Teacher answers & working

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E3

The density of X is f(x) = $\frac{1}{50}$x for 0 ≤ x ≤ 10, and 0 otherwise.Find the upper quartile of X.

Answer:

8.660

Working:

Use the CDF, because quartiles are found from cumulative probability.

The density integrates to F(x) = ∫ from 0 to x of ($\frac{1}{50}$)t dt = $\frac{x^2}{100}$ for 0 ≤ x ≤ 10.

The upper quartile is the value q where F(q) = 0.75.

$\frac{q^2}{100} = 0.75$.

$q^2 = 75$.

q = sqrt(75) ≈ 8.660.

upper quartile ≈ 8.660